A 2-ADIC NONEXISTENCE CONDITION FOR STRONGLY REGULAR GRAPHS

DILAN DJEMATENE

Abstract. For a strongly regular graph with integral eigenvalues $k > r > s$ such that $r-s$ is odd, we show that the integral eigenvalue lattices $\Lambda_\theta = \mathbb{Z}^v \cap \ker(A-\theta I)$ are even lattices of odd determinant under a mild 2-adic hypothesis, and we determine the prime-to-$(r-s)$ part of their determinants. Since an even lattice of odd determinant and rank m has determinant $\equiv (-1)^{m/2} \pmod 4$, this yields a simple arithmetic feasibility condition. It rules out 35 parameter sets with $v \le 1300$ that are listed as open in Brouwer's table of strongly regular graphs, the infinite family of negative Latin square type parameters $\mathrm{NL}_e(n)$ with e odd and all prime divisors of n congruent to 1 modulo 4, and the partial geometry $\mathrm{pg}(14,14,4)$. It also gives a five-line proof of the nonexistence of $\mathrm{srg}(289,54,1,12)$, due to Bondarenko and Radchenko.

1. Introduction

A strongly regular graph $\mathrm{srg}(v,k,\lambda,\mu)$ must satisfy a number of classical feasibility conditions: integrality of the eigenvalue multiplicities, the Krein conditions, the absolute bound, and several more specialised conditions collected in [3]. Brouwer's online table [2] records, for every feasible parameter set with $v \le 1300$, whether a graph is known to exist, is known not to exist, or is open.

In this note we prove a condition that comes from the arithmetic of the *integral* eigenvalue lattices of the adjacency matrix. Its proof only uses standard facts about even lattices over the 2-adic integers and about discriminant groups. Nevertheless, it decides 35 open entries of the table (Table 1) and an infinite family (Corollary 3).

2. Setting and statement

Let Γ be an $\mathrm{srg}(v,k,\lambda,\mu)$ with adjacency matrix A and integral eigenvalues $k > r > s$, of multiplicities $1, f, g$. Put $d = r-s$. For $\theta \in \{r,s\}$ let θ' be the other one, m_θ the multiplicity of θ, $V_\theta = \ker(A-\theta I) \subseteq \mathbb{Q}^v$, and

$$\Lambda_\theta = \mathbb{Z}^v \cap V_\theta,$$

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a primitive sublattice of the standard lattice $\mathbb{Z}^v$ of rank m_θ. For a prime q we write v_q for the q-adic valuation. The orthogonal projection onto V_θ is

$$E_\theta = \frac{A-\theta' I}{\theta-\theta'} - \frac{k-\theta'}{(\theta-\theta')\,v}\,J. \tag{1}$$

Theorem 1. *Let $\theta \in \{r,s\}$ and assume that d is odd and $v_2(k-\theta') \geq v_2(v)$ (automatic when v is odd). Then:*

(a) *Λ_θ is an even lattice of odd determinant. Consequently m_θ is even and*
$$(-1)^{m_\theta/2} \det \Lambda_\theta \equiv 1 \pmod 4.$$
(b) *For every prime $q \nmid d$ we have $v_q(\det \Lambda_\theta) = v_q(v)$ if $v_q(k-\theta') < v_q(v)$, and $v_q(\det \Lambda_\theta) = 0$ otherwise.*

In particular $\det \Lambda_\theta = \prod_{q\mid d} q^{c_q} \cdot P_\theta$ for some integers $c_q \geq 0$, where

$$P_\theta = \prod_{\substack{q\mid v,\ q\nmid d \\ v_q(k-\theta')<v_q(v)}} q^{v_q(v)}.$$

Corollary 2. *Under the hypotheses of Theorem 1, if every prime divisor of d is congruent to 1 modulo 4, then*

$$(-1)^{m_\theta/2}\, P_\theta \equiv 1 \pmod 4.$$

If this congruence fails for some θ, there is no $\operatorname{srg}(v,k,\lambda,\mu)$.

Proof. By Theorem 1, $\det \Lambda_\theta = \prod_{q\mid d} q^{c_q} P_\theta$, and the first factor is $\equiv 1 \pmod 4$. □

3. Proof of Theorem 1

We use three standard facts.

(F1) If L is a primitive sublattice of $\mathbb{Z}^v$, the orthogonal projection of $\mathbb{Z}^v$ onto $L \otimes \mathbb{Q}$ is the dual lattice $L^\#$. Indeed, it lies in $L^\#$, and every integral linear form on L extends to $\mathbb{Z}^v$ because L is primitive, and is then given by an element of the unimodular lattice $\mathbb{Z}^v$.

(F2) If L is primitive in a unimodular lattice, then L and $L^\perp$ have isomorphic discriminant groups ([5, Prop. 1.6.1]). Both groups are isomorphic to $M/(L \oplus L^\perp)$, where M is the unimodular lattice.

(F3) Let L be an even lattice over $\mathbb{Z}_2$ whose Gram determinant is a unit. Then L has even rank m and its determinant is $\equiv (-1)^{m/2} \pmod 4$. Indeed, the bilinear form is nondegenerate modulo 2 and alternating there (as L is even), so there are $x, y \in L$ with (x,y) odd. The Gram matrix of $\mathbb{Z}_2 x + \mathbb{Z}_2 y$ is $\begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$ with b odd. Its determinant $4ac - b^2 \equiv -1 \pmod 4$ is a unit, so this plane splits off orthogonally, and induction gives the claim (compare [4, Ch. 15]). A change of $\mathbb{Z}_2$-basis multiplies the determinant by a square unit, which is $\equiv 1 \pmod 8$, so the congruence holds in any basis.

Evenness. Since $\theta \neq k$, $V_\theta \perp \mathbf{1}$, so every $x \in \Lambda_\theta$ has $\sum x_i = 0$. Hence $x \cdot x = \sum x_i^2 \equiv \sum x_i \equiv 0 \pmod 2$.

Odd determinant. The first term of (1) has denominator $\pm d$, which is odd. The second is 2-integral because $v_2(k-\theta') \geq v_2(v)$. So E_θ has 2-integral entries. By (F1), $\Lambda_\theta^\# \otimes \mathbb{Z}_2 = E_\theta(\mathbb{Z}_2^v) \subseteq \mathbb{Z}_2^v \cap V_\theta = \Lambda_\theta \otimes \mathbb{Z}_2$, so Λ_θ is unimodular at 2, that is, $\det \Lambda_\theta$ is odd. Part (a) now follows from (F3), applied to $\Lambda_\theta \otimes \mathbb{Z}_2$ in a $\mathbb{Z}$-basis of Λ_θ.

Part (b). Let $q \nmid d$. If $v_q(k-\theta') \geq v_q(v)$, then E_θ is q-integral, and the argument above with q in place of 2 shows $q \nmid \det \Lambda_\theta$. Assume now $v_q(k-\theta') < v_q(v)$, so $q \mid v$. Let $W = \mathbb{Z}^v \cap \mathbf{1}^\perp$. By (F2), $\operatorname{disc} W \cong \operatorname{disc}(\mathbb{Z}\mathbf{1}) \cong \mathbb{Z}/v$ is cyclic. On W the matrix J vanishes, so $E_\theta|_W = (A - \theta' I)/(\theta - \theta')$ is q-integral. Since $\Lambda_\theta, \Lambda_{\theta'} \subseteq W$ are primitive, we get an orthogonal decomposition

$$W \otimes \mathbb{Z}_q = (\Lambda_\theta \otimes \mathbb{Z}_q) \perp (\Lambda_{\theta'} \otimes \mathbb{Z}_q),$$

and therefore the cyclic group $(\mathbb{Z}/v) \otimes \mathbb{Z}_q \cong \mathbb{Z}/q^{v_q(v)}$ is the direct sum of the q-parts of $\operatorname{disc} \Lambda_\theta$ and $\operatorname{disc} \Lambda_{\theta'}$. A cyclic q-group is not a direct sum of two nontrivial groups, so one of these q-parts is trivial and the other is $\mathbb{Z}/q^{v_q(v)}$.

It remains to see which one is trivial. From $(k-r)(k-s) = \mu v$ and $(k-s)-(k-r) = d$ with $q \nmid d$, the prime q divides exactly one of $k-r$, $k-s$, and that factor has valuation $v_q(\mu v) \geq v_q(v)$. By assumption it is not $k-\theta'$, so $v_q(k-\theta) \geq v_q(v)$. Then $E_{\theta'}$ is q-integral and $\Lambda_{\theta'}$ is q-unimodular. Hence $v_q(\det \Lambda_\theta) = v_q(v)$. □

4. Consequences

Corollary 3 (Negative Latin square type)**.** *Let n be an integer all of whose prime divisors are $\equiv 1 \pmod 4$, and let e be odd. Then there is no strongly regular graph with the parameters*

$$\mathrm{NL}_e(n) = \operatorname{srg}\bigl(n^2,\ e(n+1),\ -n+e^2+3e,\ e^2+e\bigr).$$

Proof. The eigenvalues are $r = e$ and $s = e - n$, so $d = n$. Since $v = n^2$ is odd, every prime of v divides d, and $P_s = 1$. The multiplicity of s is $m_s = e(n+1)$ (from $\operatorname{tr} A = 0$). As e is odd and $n \equiv 1 \pmod 4$, $m_s \equiv 2 \pmod 4$, so $(-1)^{m_s/2} P_s = -1$, contradicting Corollary 2. □

For example, $\mathrm{NL}_3(17) = \operatorname{srg}(289, 54, 1, 12)$, whose nonexistence was proved by Bondarenko and Radchenko [1] by other means. The table contains open members of this family for $n \in \{13, 17, 25, 29\}$, see Table 1. At $v = 625$, the table records graphs with the parameters $\mathrm{NL}_e(25)$ for $e = 4, 6, 8, 10, 12$, while $e = 5, 7, 9, 11$ were open; this matches Corollary 3.

Corollary 4. *There is no partial geometry* $\operatorname{pg}(14, 14, 4)$.

Proof. Its point graph would be an $\operatorname{srg}(750, 210, 55, 60)$, with eigenvalues $r = 10$, $s = -15$, $d = 25$ and $m_s = 308$. Take $\theta = s$: $k - r = 200$, so $v_2(k-r) = 3 \geq v_2(750) = 1$. The primes of $750 = 2 \cdot 3 \cdot 5^3$ not dividing d

are 2 and 3. We have $v_2(200) \geq v_2(750)$ and $v_3(200) = 0 < 1$, so $P_s = 3$. Then $(-1)^{154} \cdot 3 = 3 \not\equiv 1 \pmod 4$. □

TABLE 1. Parameter sets with $v \leq 1300$ marked open ("?") in Brouwer's table (snapshot of September 2026) that are excluded by Corollary 2. The column θ gives the eigenvalue used, with its multiplicity m and P_θ.

(v,k,λ,μ)	θ	m	P_θ	(v,k,λ,μ)	θ	m	P_θ
(169,42,5,12)	3	126	1	(630,68,1,8)	5	440	63
(169,70,27,30)	5	98	1	(690,318,121,168)	3	644	15
(209,52,15,12)	8	76	11	(721,220,69,66)	14	308	103
(235,52,9,12)	5	140	47	(750,210,55,60)	−15	308	3
(289,90,23,30)	5	198	1	(750,214,63,60)	−11	428	3
(289,126,53,56)	7	162	1	(780,369,158,189)	−36	104	15
(375,154,53,70)	4	308	3	(841,150,11,30)	5	690	1
(375,170,85,70)	20	68	3	(841,210,41,56)	7	630	1
(391,140,39,56)	4	322	17	(841,270,79,90)	9	570	1
(430,39,8,3)	−4	300	43	(841,330,125,132)	11	510	1
(441,190,89,76)	19	98	9	(841,390,179,182)	13	450	1
(494,85,12,15)	−10	208	19	(871,348,137,140)	13	468	67
(507,44,1,4)	5	308	3	(875,184,33,40)	9	552	7
(616,205,90,57)	−4	560	11	(931,180,29,36)	9	588	19
(625,130,15,30)	5	494	1	(1026,451,212,187)	−8	836	27
(625,182,45,56)	7	442	1	(1148,481,210,195)	26	328	7
(625,234,83,90)	9	390	1	(1248,203,22,35)	8	896	39
(625,286,129,132)	11	338	1				

5. Verification and remarks

Remark 5 (Consistency with known graphs). The condition was implemented twice, independently; one implementation is the accompanying file `verify_s5.py`. Both were applied to all 4538 rows of Brouwer's table with $v \leq 1300$, complements included. No parameter set for which a graph is known is excluded. Among the parameter sets known not to exist, the condition recovers $(289, 54, 1, 12)$ [1] and $(154, 51, 8, 21)$, which the table excludes by the Krein condition.

Remark 6 (Scope). When d has a prime divisor $q \equiv 3 \pmod 4$, the exponent c_q is not controlled. Theorem 1 then only gives the parity of m_θ, which in practice is already forced by integrality. This is the case of Conway's 99-graph problem, $\mathrm{srg}(99, 14, 1, 2)$, where $d = 7$. The hypothesis $v_2(k - \theta') \geq v_2(v)$ cannot be dropped when v is even. For the Petersen graph, Λ_1 has determinant 162.

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